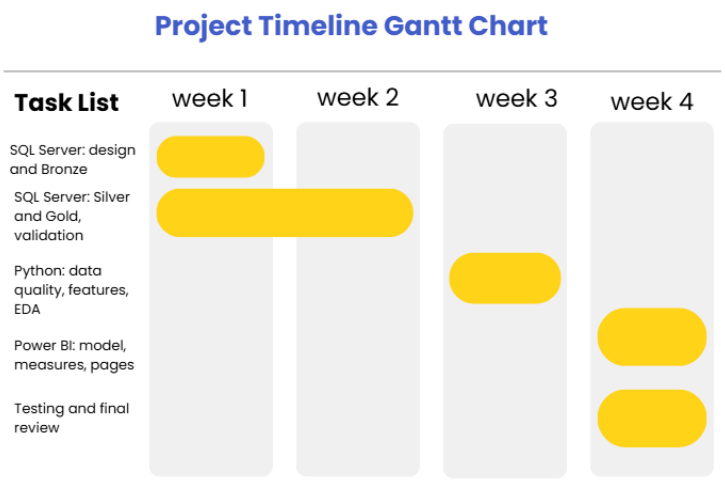
Project Planning & Management

1. Project Plan

Timeline (Gantt chart)



Milestones and deliverables

Milestone	End of	Deliverable
M1: Warehouse designed and raw data loaded	Week 1	Schema design, Bronze tables loaded and validated, SQL validation queries
M2: Warehouse complete	Week 2	Cleaned Silver data, Gold fact and dimension tables, data quality review, business queries
M3: Analysis complete	Week 3	Data quality fixes, derived columns, EDA notebooks, business insights
M4: Dashboard complete	Week 4	Data model, DAX measures and KPIs, Executive Overview page, analytical pages

Resource allocation

- **People:** 5 team members, with every member working in all three phases.
- **Tools:** SQL Server, Python, and Power BI.
- **Data:** a fixed snapshot of the IMF PortWatch data, with the download date recorded.
- **Time:** 4 weeks, with a weekly team check-in at the end of each milestone.

2. Task Assignment & Roles

Member	Role	Responsibilities
1	Data Architect	SQL: schema design, relationships, part of Gold. Python: data quality checks. Power BI: data modeling and filter propagation testing
2	Data Loading & Measures Lead	SQL: loading data into Bronze and validation. Python: feature engineering. Power BI: DAX measures, KPIs, and checking the calculations
3	Data Cleaning & Overview Lead	SQL: cleaning in Silver (missing values, duplicates). Python: univariate EDA. Power BI: Executive Overview page
4	Warehouse & Analysis Lead	SQL: fact and dimension tables in Gold. Python: bivariate and multivariate EDA. Power BI: analytical page with drill-down and filters
5	Quality & Insights Lead	SQL: validation queries, data quality review, business queries. Python: EDA on an independent part of the data and business insights. Power BI: analytical page and testing interactions and consistency of results

3. Risk Assessment & Mitigation Plan

Risk	Likelihood / Impact	Mitigation
PortWatch updates or revises its data during the project	Medium / High	Use one fixed snapshot, record the download date, and don't mix versions
Cargo values are preliminary estimates	High / Medium	Label them as estimates, and report traffic and estimated volume, not economic loss
Disruption events don't link cleanly to chokepoints	Medium / Medium	Check the link early in Week 1, and define the main events manually with sources if needed
A phase depends on the previous one and is delayed	Medium / High	Weekly check-in, start Silver in Week 1, and keep tasks small and clearly owned
A member is unavailable or the workload is uneven	Medium / Medium	Document every task, review each other's work, and rebalance in the weekly check-in
Dashboard numbers don't match the SQL results	Medium / High	Validation queries in each layer, and Member 5 tests that results are consistent

Scope creep (prices, forecasting)	Medium / Medium	Keep the “Won’t have” list from the requirements and approve changes through the change log

4. KPIs

KPI	**Target**
Data coverage	All 28 chokepoints loaded, with row counts matching the source
Data quality	All SQL validation queries pass, and no duplicates remain in Silver
Accuracy	Dashboard numbers match the SQL results for the checked measures
Dashboard response time	Filters and drill-down update in about 3 seconds or less
On-time delivery	All 4 milestones delivered by the end of their week
Team participation	All 5 members complete their tasks in all 3 phases
Usefulness	The dashboard answers the project’s business questions (impact, recovery, and spillover), and the lecturer approves it